



[Person]

Views processed or raw energy data,
correlations in form of charts,
feeds provided by the SLM



Energy Correlation Detection System

Web Application

[Container, TypeScript and Angular]

Provides charts, tables, and a feed of
interpretations via an SLM

Makes API calls
[JSON/HTTPS]

Sends real-time
energy data
[WebSockets, HTTPS]

Backend/API

[Container, Spring Boot and Kafka]

Provides raw and processed energy data via
JSON/HTTPS

Reads from
and writes to
[ORM]

Database

[Container, Relational Database]

Stores raw energy data, and processed data
on correlations

Real-time Event Stream

[Container, Apache Kafka]

Buffers and distributes energy events
between system components

consumes energy
events
[JSON,
Kafka protocol]

publishes energy
events
[JSON,
Kafka protocol]

consumes derived
correlation events
[JSON, Kafka protocol]

publishes correlation
interpretation events
[JSON, Kafka protocol]

Energy Data Interpreter

[Container, Ollama]

Transforms processed energy data into
plain text interpretations

Carbon Intensity API

[External System]

Provides real-time data on carbon intensity
regarding different energy sources

Weather API

[External System]

Provides real-time data on weather
phenomena, such as wind drought, cloud
coverage, etc.

Energy Grid API

[External System]

Provides real-time data on the energy grid for
categories such as load, generation, etc.

fetches
data

